UNIVERSITY OF CENTRAL PUNJAB LAHORE

Assignment # 2 Version 1

CLO Mapping: CLO2

Course Code	**MAT243**	**Semester**	**Fall 2025**
Course Title	**Multivariable Calculus**		
Resource Person	**Dr. Muhammad Rayees Ahmad**		
Assignment Given Date	**3-11-2025**	**Submission Date**	**10-11-2025**
Total Marks	**35= Assignment (30) + Quiz based Viva (5)**		

Version	Name of student	Complete Registration no.	Section	Assignment marks	Viva marks	Total marks
1						
2						

Submission Instructions *(Please follow strictly)*

- **Take the PRINTOUT OF THIS ASSIGNMENT PDF file and solve the question at the given empty spaces in the assignment PDF file. DO NOT use any other papers. Alternatively, you can only use simple plane white printer pages.**

- **This is an INDIVIDUAL ASSIGNMENT.**
- **Version 1 will solve the students who has the last digit ODD in their registration number.**
- **Version 2 will solve the students who has the last digit EVEN in their registration number.**
- **It is mandatory to know the whole method of all versions. Any question may be asked in the viva.**

- **Write your complete names and registration numbers on each page of the assignment.**
- **Follow the deadline. Finish your work one day before, so you could submit in time.**

- **LATE SUBMISSION WILL NOT BE ACCEPTED (Deduction of 5% marks each day).**
- **All Plagiarized assignments will be awarded ZERO marks.**

I declare that I have prepared the assignment according to above guidelines, and I shall be responsible for any deduction of marks if the instructions are not followed.

Student Signature: _____

Name: **Registration No.**

Version: 1

Question [CLO2] (30 marks):

In a **robotic navigation system**, the motion of a drone in three-dimensional space is represented by a **vector-valued velocity function**:

$$\vec{v}(t) = < \ln(t+1),\ \frac{1}{2t-6},\ \sin(3t-1) >$$

where t (in seconds) represents time since launch, and each component corresponds to the instantaneous velocity of the drone along the x-axis, y-axis, and z-axis, respectively.

The **initial position** of the drone at $t = 0$ is given by

$$\vec{r}(0) = < 2, 1, 0 >$$

a) **Find** the domain of the velocity vector $\vec{v}(t)$ and plot the **solution space** on real number line.

b) **Determine** the **acceleration vector** $\vec{a}(t)$ of the drone using the given **velocity** function.

c) **Determine** the **position vector** $\vec{r}(t)$ of the drone using the given **velocity** function.

d) Using the given initial condition, **find** the complete expression for $\vec{r}(t)$.

e) **Compute** the position of the drone at $t = 1$ second.

Marks Distribution: **part (a): 15 marks, part (b): 5 marks, part (c): 6 marks, part (d): 2 marks, part (e): 2 marks.**

Solution:

Name: **Registration No.**

Name: **Registration No.**

UNIVERSITY OF CENTRAL PUNJAB LAHORE

Assignment # 2 Version 2

CLO Mapping: CLO2

Course Code	**MAT243**	**Semester**	**Fall 2025**
Course Title	**Multivariable Calculus**		
Resource Person	**Dr. Muhammad Rayees Ahmad**		
Assignment Given Date	**31-10-2025**	**Submission Date**	**28-10-2025**
Total Marks	**35= Assignment (30) + Quiz based Viva (5)**		

Version	Name of student	Complete Registration no.	Section	Assignment marks	Viva marks	Total marks
1						
2						

Submission Instructions *(Please follow strictly)*

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- **Write your complete names and registration numbers on each page of the assignment.**
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- **All Plagiarized assignments will be awarded ZERO marks.**

I declare that I have prepared the assignment according to above guidelines, and I shall be responsible for any deduction of marks if the instructions are not followed.

Student Signature: _____

Name: **Registration No.**

Version: 2

Question [CLO2] (30 marks):

In an **autonomous vehicle system**, a self-driving car moves along a 3D test track where its motion is represented by a **vector-valued velocity function**:

$$\vec{v}(t) = < \ln(3t),\ \sqrt{16 - t^2},\ \cos(2t) >$$

where t (in seconds) represents the time since the start of the test, and each component corresponds to the instantaneous velocity of the car along the x-axis, y-axis, and z-axis, respectively.

The **initial position** of the vehicle at $t = 0$ is given by

$$\vec{r}(0) = < 3, 2, 0 >$$

a) **Find** the **domain** of the **velocity vector** $\vec{v}(t)$ and plot the **solution space** on real number line.

b) **Determine** the **acceleration vector** $\vec{a}(t)$ of the car using the given **velocity** function.

c) **Determine** the **position vector** $\vec{r}(t)$ of the car using the given **velocity** function.

d) Using the given initial condition, **find** the complete expression for $\vec{r}(t)$.

e) **Compute** the position of the car at $t = 1$ second.

Marks Distribution: **part (a): 15 marks, part (b): 5 marks, part (c): 6 marks, part (d): 2 marks, part (e): 2 marks.**

Solution:

Name: **Registration No.**

Name: **Registration No.**